

# Ziyu (Rick) Xu

Robotics Institute, Carnegie Mellon University

**Research Interests:** Vision-Language Models, Streaming Video, and Efficient Inference

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## Education

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| <b>Carnegie Mellon University (CMU)</b>                                                                 | <b>Aug. 2026 – Expected 2028</b> |
| Master of Science in Robotics (MSR)                                                                     | Pittsburgh, PA                   |
| <b>University of Michigan, Ann Arbor (UM)</b>                                                           | <b>Oct. 2024 – May 2026</b>      |
| B.S.E. in Mechanical Engineering (Robotics Track); Minor in Computer Science                            | GPA: 3.85/4.0 (Top 5%)           |
| • <b>Core Coursework:</b> Robot Learning, RL, Embedded Systems, Optimal Control, Convex Optimization    |                                  |
| <b>Shanghai Jiao Tong University (SJTU)</b>                                                             | <b>Sep. 2022 – Jul. 2026</b>     |
| B.S.E. in Electrical and Computer Engineering                                                           | GPA: 3.56/4.0 (Top 20%)          |
| • <b>Core Coursework:</b> Algorithms, Mathematical Foundations, Probability Theory, Signals and Systems |                                  |

## Research Experiences and Internships

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| <b>Shanghai Artificial Intelligence Lab &amp; IRMV, SJTU</b>                                                                                                                                                                                                                                                                   | <b>Apr. 2026 – Present</b>   |
| Research Assistant, Directed by Prof. Junchi Yan and Prof. Hesheng Wang                                                                                                                                                                                                                                                        | Shanghai, China              |
| • <b>Iterative Visual KV Compression:</b> Studied errors from reusing retained visual KV entries after neighboring tokens are removed in multi-turn and streaming inference. Developed context-aware re-encoding at original positions and compared with inheritance and restoration under matched memory and runtime budgets. |                              |
| <b>Unitree Robotics, R&amp;D Department, Embodied AI Team</b>                                                                                                                                                                                                                                                                  | <b>May 2026 – Present</b>    |
| Research Intern, Humanoid Robot Foundation Models                                                                                                                                                                                                                                                                              | Hangzhou, China              |
| • <b>VLA Adaptation and Post-Training:</b> Deployed and fine-tuned Isaac GR00T/OpenPI policies with SONIC whole-body control using real-robot rollout traces. Developing phase-aware adaptation across embodiments by aligning semantic task progress and execution timing.                                                    |                              |
| • <b>Reproducible Evaluation:</b> Implemented an 8-task, 24-scene manipulation benchmark with background variation. Built PICO-based scene digitization and AR-guided reconstruction with reusable layouts, achieving 2.5 cm median placement error.                                                                           |                              |
| <b>Surgical Computer Vision Detection, SJTU</b>                                                                                                                                                                                                                                                                                | <b>Apr. 2024 – Dec. 2024</b> |
| Research Assistant, Directed by Prof. Yutong Ban                                                                                                                                                                                                                                                                               | Shanghai, China              |
| • <b>Surgical Video Segmentation:</b> Evaluated Segment Anything Model variants on surgical frames, examining domain shift and temporal mask stability around instruments, tissues, and interaction regions.                                                                                                                   |                              |

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## Intelligent Control Lab, Carnegie Mellon University

Graduate Student Researcher, advised by Prof. Changliu Liu

Research experience continued

**Aug. 2026 – Present**

Pittsburgh, PA

- **Whole-Body and Safe Locomotion Research (ongoing):** Investigating whole-body control, safety with control barrier functions (CBFs), and world-model-based locomotion.
- **Dexmate Bimanual Manipulation Demo, IROS 2026:** Contributed to a bimanual block-stacking demonstration combining agent planning with policy execution. Implemented the agent component, prepared the on-site environment, adapted the system to different grippers, and performed system calibration.

## Computational Autonomy and Robotics Lab, UM

**Jun. 2025 – Apr. 2026**

Research Assistant, Directed by Prof. Maani Ghaffari

Ann Arbor, MI

- **Field Perception:** Built DINO/SAM2 segmentation and fine-tuned Grounded-DINO/YOLO for submerged vegetation detection in autonomy experiments.

## Intelligent Robotics and Autonomy Lab, UM

**Oct. 2024 – Apr. 2026**

Research Assistant, Directed by Prof. Vasileios Tzoumas

Ann Arbor, MI

- **Multi-Robot Simulation and ROS2:** Rebuilt a Unity + AirSim multi-drone testbed with revised dynamics and sensing; migrated SlideSLAM communication from ROS1 to ROS2 for bandwidth-limited active-SLAM evaluation.

## Publications and Manuscripts

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- Xiaohan Wang\*, **Ziyu Xu\***, Xuyi Yang. “Re-encoding What You Keep After Every Compression: Retained Visual KV Is Computed in a Context That No Longer Exists”, *Submitted to ICLR 2027*. \*Equal contribution.
- Xuyi Yang, Xiaohan Wang, Wenhao Zhang, and **Ziyu Xu**. “StreamJIT: Continuous Observation and Just-in-Time Deliberation for Proactive Streaming Video Assistants”, *Submitted to ICLR 2027*.
- C. Yuan, J. Jiang, K. Yang, **Z. Xu**, . . . , and Y. Ban. “Systematic Evaluation and Guidelines for Segment Anything Model in Surgical Video Analysis”, *npj Digital Surgery*, accepted, 2024.
- Yinan Dong, **Ziyu Xu**, Tsimafei Lazouski, Sangli Teng, and Maani Ghaffari. “Online Learning-Enhanced Lie Algebraic MPC for Robust Autonomous Surface Vehicle Control”, *IEEE Robotics and Automation Letters (RA-L)*, revised manuscript under review, 2025.

## Selected Awards

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Gold Medal, University Physics Competition (2023); Dean’s Honor List, University of Michigan (2024, 2025).

## Skills

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- **Multimodal ML:** PyTorch, visual KV compression, streaming/multi-turn inference evaluation, LLM fine-tuning, SAM2, Grounded-DINO, YOLO, OpenCV
- **Systems:** Python/C++, Linux, Git, Docker, ROS/ROS2, real-robot policy deployment